Ibuprofen

Mechanism of action

- Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) that is predominantly used for treating pain.
- Ibuprofen inhibits the activity of the enzyme prostaglandin synthetase, reducing prostaglandin production and causing a reduction in inflammation, pain and fever.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to 'pain relief' in the FR Field Guide

Indications

- ✓ Mild to moderate pain, usually in combination with paracetamol.
- ✓ Moderate to severe pain, usually in combination with other medicines.

Contraindications

- ✗ Known severe allergy.
- ✗ Pregnancy.
- ✗ The presence of sepsis, dehydration, shock or clinically significant bleeding. Ibuprofen can worsen renal impairment and increase the risk of bleeding.
- ✗ Known worsening of bronchospasm with NSAIDs. Some patients with asthma or COPD have known worsening of bronchospasm with NSAIDs. If there is a clear history of significant bronchospasm with NSAIDs, ibuprofen must not be administered.

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Cautions

- The patient has taken ibuprofen within the last four hours. Ibuprofen is contained in many products such as cold and flu tablets/drinks, combination analgesics and migraine tablets. Additional ibuprofen may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold ibuprofen if there is any doubt.
- Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- Age greater than or equal to 75 years, particularly if frail. In this setting, renal impairment is likely and ibuprofen may worsen renal impairment.

Use in pregnancy or when breastfeeding

- May cause foetal harm during pregnancy and should not be administered. In particular, ibuprofen has been associated with premature delivery and premature closure of the ductus arteriosus when administered during the third trimester of pregnancy.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 600 mg for an adult weighing greater than 80 kg.
- 400 mg for an adult weighing 80 kg or less.
- See the paediatric drug dose tables for a child.

Administration

- Administer PO.
- Children unable to swallow tablets may be administered ibuprofen syrup, or tablets that have been crushed and placed in a soft food such as jam or honey.

Common adverse effects

- Renal impairment.

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- Increased bleeding.
- Although indigestion, gastrointestinal ulceration and gastrointestinal bleeding are commonly listed as adverse effects, these are only associated with chronic administration.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 4-6 hours.

Usual preparation

- 200 mg tablets.
- Syrup containing 20 mg/ml.

Pharmacokinetics

- Ibuprofen is absorbed in the stomach and small intestine. The presence of food in the stomach will delay absorption, but this is not usually clinically significant.
- Ibuprofen is metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Warfarin. Ibuprofen displaces warfarin from binding sites and increases the activity of warfarin.

Additional information

- Ibuprofen is not indicated for pain associated with myocardial ischaemia.
- All personnel (including those without ATP) may give ibuprofen to a patient for self-administration, provided the package instructions are followed.

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- Ibuprofen is useful in the context of soft tissue pain, musculoskeletal pain and renal colic, particularly if an opiate is not going to be administered.
- Ibuprofen may be administered in addition to other medicines for mild to moderate pain, particularly those not requiring transport or opiate analgesia.
- A patient may be administered ibuprofen for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required.

More Information

Version history

22 November 2022 1.0.1

Update: Indications, contraindications and cautions.